

VERITYFLOW

Trusted evidence. Controlled intelligence.

Product Requirements Document

YC Fall 2026 x Moss The Zero Latency Builder Sprint

Draft 0.1 · September 2026

Product Summary

VerityFlow is a reporting agent based on evidence which converts validated operational data into reliable and context-aware reports; deterministic engines are used to calculate facts and detect findings, Moss obtains the appropriate business context, and an AI agent prepares narratives that can be reviewed. Analysts still have final control as to what is approved and published.

Problem Statement

VerityFlow was developed as a result of the problems that arose when dealing with renewable-energy data and as a response to the frequent solar-performance reports. Preparing these reports required much more than just writing a summary or producing charts; analysts had to repeatedly check the quality of the operational data, spot any missing or invalid observations, reconcile the column names specific to each customer, apply the approved KPI formulas, investigate any performance exceptions, answer the usual questions raised by stakeholders, and explain the evidence supporting each conclusion before the reports could be prepared. A great deal of the work relied on the knowledge possessed by individual analysts—such as how a customer defines a KPI, what aggregation method they anticipate, which thresholds count as a significant finding, what kind of evidence should support an explanation, and what information should be included in the final report. This knowledge was typically spread out over spreadsheets, reporting templates, business documents and earlier reports. Although existing reporting tools are able to calculate and visualise data they do not consistently retain and apply the approved analytical context. While generic AI tools can generate fluent narratives, those narratives may be incomplete, misleading or hard to verify since they lack governed evidence and knowledge specific to the customer. By combining deterministic calculations with approved business context, evidence-based AI explanations and a review by an analyst within a single controlled workflow, VerityFlow deals with this gap. Although the product first emerged in the area of renewable-energy reporting, the same problem also occurs in finance, operations, compliance and in other recurring reporting environments where accuracy, context and accountability are important.

Product Approach

VerityFlow deals with this issue by means of a controlled reporting process which integrates deterministic calculations, approved business context, Moss-driven retrieval, AI-generated narratives that are based on evidence, and a step involving analyst review. It turns validated operational data into draft reports that can be traced while at the same time making sure that the analysts keep control over what is edited, approved, and published.

Product vision

The first VerityFlow agent will show how governed context retrieval can be used for recurring solar-performance reports. The same workflow can subsequently be adapted for use with finance, operations, compliance, and other types of recurring reports where conclusions have to be supported by verifiable evidence and human review.

Target Users

Primary user: Reporting analyst

- Configures mappings, formulas and rules.
- Reviews data-quality issues.
- Approves explanations and final reports.

Secondary user: Operations or business manager

- Receives the report.
- Needs concise, trustworthy explanations.
- Wants to understand what requires attention.

Customer or stakeholder

- Receives the approved PDF.
 - Needs reliable performance information without technical clutter
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Product Goals

- Generate narratives grounded in deterministic evidence.
- Retrieve only relevant, approved customer context.
- Create reusable report output.
- Preserve analytical knowledge.
- Keep analyst control over publication.
- Support multiple data source.
- Make every narrative traceable to evidence and retrieved context.

What VerityFlow is not

- Not a data-cleaning product.
 - Not a generic RAG chatbot.
 - Not an AI KPI calculator.
 - Not an autonomous publishing system without approval.
 - Not a replacement for BI visualisation platforms.
 - Raw operational data will not be stored in Moss for calculation.
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Current Capabilities

Data input and validation

- Accepts clean inputs from Excel and BigQuery
- Identifies missing sheets, columns, invalid values, date gaps, and other data-quality issues if there are any.
- Gives analysts meaningful correction guidance without automatically modifying the source data.

Governed analytical configuration

- Stores customer, site, capacity, branding, and report configuration.
- Maps customer-specific column names to standard metrics.
- Calculates KPIs using deterministic, analyst-approved formulas.
- Allows customer-specific aggregation methods to be configured and reused.
- Applies approved business rules to identify significant findings.
- Stores recurring customer questions and reporting preferences.

Report Builder and Review

- Produces KPI scorecards, charts, tables, heatmaps, dual-axis charts, stacked bars, and waterfall charts.
- Validates whether the selected data is suitable for the requested chart.
- Explains why an incompatible chart cannot be generated and recommends alternatives.
- Generates evidence-backed chart explanations.
- Allows analysts to edit, approve, reject, reorder, include, or exclude report components.
- Supports executive-summary placement, narrative placement, report themes, and customer/company branding.
- Saves and versions approved layouts by customer and report type.
- Creates an immutable approved-report snapshot.
- Provides customer preview and browser-generated PDF export.

Current Limitations

- Live Moss Cloud retrieval: the adapter is implemented, but real retrieval requires Moss project credentials.

- AI narrative agent: the current version uses a deterministic grounded-narrative assembler. The LLM agent is still to be built.
 - VerityFlow records how quickly context is retrieved, but it does not yet automatically test whether Moss returned the right approved information, missed any important information, or whether every narrative claim is supported by the retrieved context and calculated evidence.
 - Context-management UI: the approved context store exists, but analysts cannot yet create and approve all context through a dedicated interface.
 - Production BigQuery pilot: the adapter exists, but it has not yet been validated with Digitised Energy's live data.
 - Scheduling and email: automatic PDF generation and scheduled customer delivery remain roadmap capabilities.
 - Hosted deployment: VerityFlow currently runs locally.
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Functional Requirement

ID	Requirements
FR_01	VerityFlow shall calculate KPI values without using an LLM.
FR_02	VerityFlow shall index only active, analyst-approved context.
FR_03	Retrieval shall be filtered by customer, report type, KPI, and approval status.
FR_04	Every generated narrative shall retain references to its evidence packet and retrieved context.
FR_05	Analysts shall be able to edit, reject, or approve generated narratives.
FR_06	Final approval shall create an immutable report snapshot and audit record.
FR_07	Context belonging to one customer shall not be retrieved for another customer.
FR_08	Raw operational data shall not be stored in Moss for KPI calculation.
FR_09	VerityFlow shall record retrieved context IDs, relevance scores, filters, and retrieval latency.
FR_10	VerityFlow shall clearly identify when fallback retrieval is used instead of Moss.
FR_11	Analyst corrections shall become reusable context only after explicit approval.
FR_12	The analyst shall be able to decide whether approved wording applies only to the current report or to future reports.
FR_13	Superseded or inactive context shall not be retrieved.
FR_14	The system shall request analyst investigation when available evidence is insufficient to support a causal explanation.

VerityFlow Primary User Journey

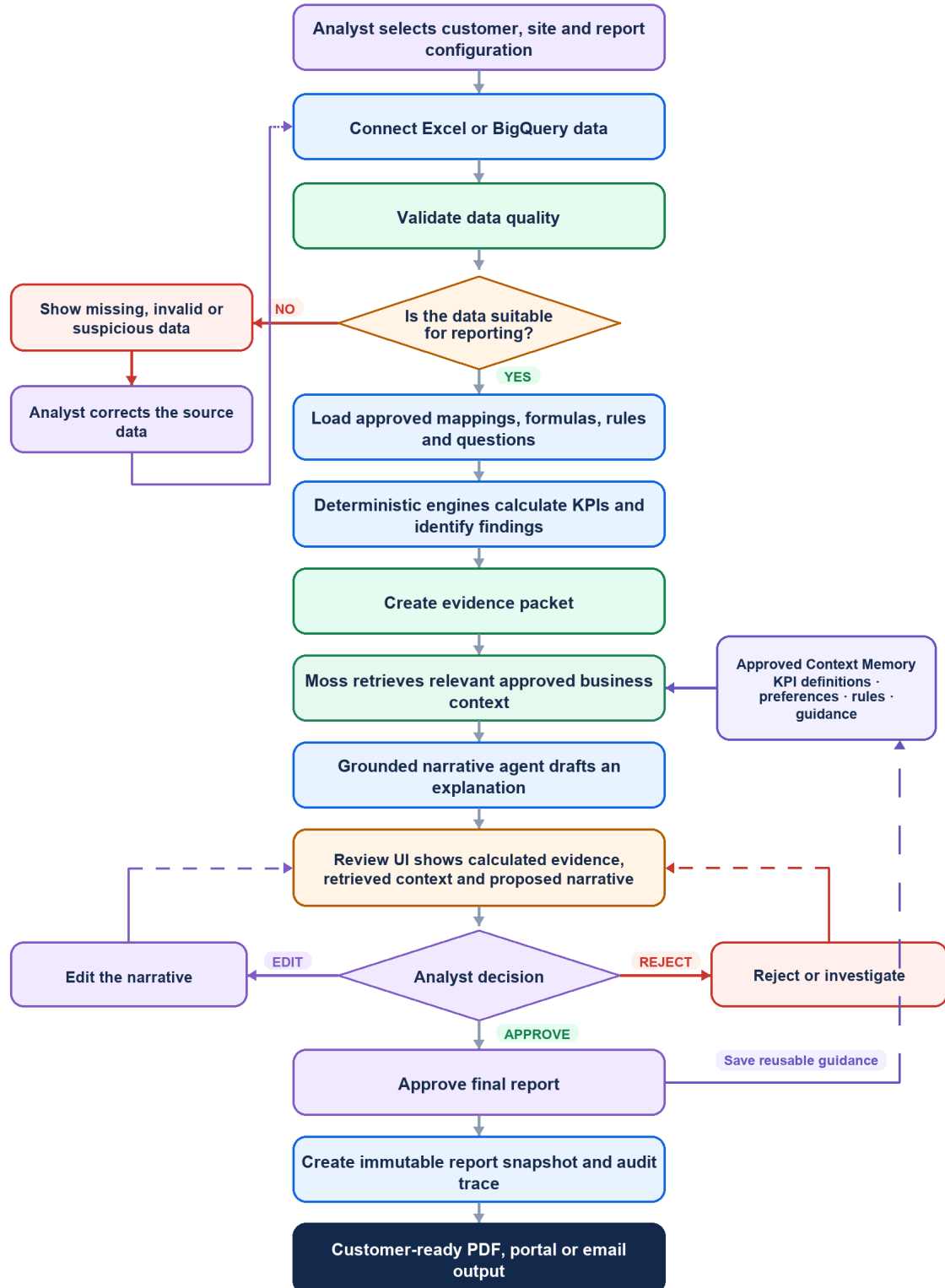
From validated operational data to an approved, context-grounded report

Analyst action

VerityFlow

Moss retrieval

Review / decision



Facts are calculated deterministically · Context is retrieved from approved memory · Analysts control publication

Moss retrieval design

Moss should index:

- Approved KPI definitions
- Customer reporting preferences
- Approved interpretation rules
- Standing customer questions
- Reporting guidelines
- Approved narrative patterns
- Explicitly saved analyst guidance

The context should contain

Field	Purpose
Context ID	Unique identifier
Customer ID	Prevents cross-customer retrieval
Report type	Limits context to the relevant report
KPI	Associates context with a metric
Context Type	Definition, rule, preference, question, or pattern
Context Text	Approved business knowledge
Approval Status	Determines whether Moss may retrieve it
Approved By	Records the responsible analyst
Version	Supports controlled updates
Source	Shows where the knowledge originated
Effective date	Identifies when the context becomes applicable
Expiry Date	Optional date after which it should not be used
Active status	Allows context to be retired without deleting history

Evidence and narrative contract

The deterministic engine supplies:

- Calculated KPI facts and units
- Triggered approved rules
- Current and historical comparisons
- Relevant dates and reporting periods
- Evidence identifiers
- Data-quality status
- Formula-supported KPI relationships
- Questions that could not be answered

The narrative agent returns:

- Draft explanation
 - Evidence identifiers used
 - Retrieved context identifiers used
 - Statements requiring analyst review
 - Unsupported or unresolved questions
 - No newly calculated KPI values
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Trust and Safety Requirements

- AI shall not generate or modify KPI values.
 - Context shall never be retrieved across customers.
 - Only active, analyst-approved context may be retrieved.
 - Causal statements require supporting formulas, rules, or evidence.
 - Possible causes must be labelled as requiring investigation.
 - Data-quality failures must remain visible.
 - Every published narrative requires analyst approval.
 - Every approval must create an audit record.
 - Fallback retrieval must not be presented as Moss retrieval.
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Success Metrics and Evaluation

Metrics	Target
Numerical claims traceable to evidence	100%
Cross-customer context leakage	0
Retrieved context that is analyst-approved	100%
Unsupported numerical claims	0
Unsupported causal claims	0
Retrieval relevance	Manually scored against expected results
Moss retrieval latency	Median and p95 recorded
Analyst outcomes	Approval, edit, rejection and investigation rates

MVP Criteria

- A deterministic finding creates an evidence packet.
- Moss retrieves the expected approved context.
- Retrieval respects customer, report, KPI, and approval filters.
- The narrative uses only the supplied evidence and retrieved context.
- The UI displays evidence and context provenance.
- The analyst can edit, reject, or approve the narrative.
- The approved result and retrieval trace are recorded.
- Retrieval latency is measured.

- Customer-isolation tests pass.
 - Narrative-grounding tests pass.
 - The complete workflow can be demonstrated within two minutes.
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Risks and Mitigation

Risk	Mitigation
Irrelevant context is retrieved	Metadata filtering, top-k limits and relevance evaluation
Approved context becomes outdated	Versioning, active status, effective and expiry dates
Cross-customer context leakage	Mandatory customer filters and isolation tests
AI makes unsupported conclusions	Evidence contract, claim validation and analyst review
Source data is incomplete	Block or qualify affected findings and show data-quality warnings
LLM is unavailable	Preserve deterministic findings and allow manual narrative review
Moss is unavailable	Clearly labelled fallback; never misrepresent fallback as Moss
Index is not refreshed	Explicit sync after approved-context changes
Too much context confuses the agent	Retrieve a small, filtered top-k context set

Dependencies

- Moss Python SDK
 - LLM provider
 - Existing FastAPI backend
 - React frontend
 - SQLite reporting memory
 - Excel and BigQuery source adapters
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Development Milestone

Milestone	Status
Approved-context schema and demo seed data	Completed
Evidence-packet creation	Completed
Moss SDK integration	Completed in code
Customer/report/KPI metadata filtering	Completed
Real Moss Cloud credential test	Pending
Review UI showing evidence and retrieved context	Completed
Retrieval latency tracing	Completed
Cross-customer isolation tests	Completed
Deterministic grounded-narrative assembly	Completed
LLM narrative agent	Pending
Retrieval relevance evaluation	Pending

Narrative-faithfulness evaluation	Pending
Two-minute hackathon demonstration	Pending